

David Manuel

Senior Software Engineer

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PROFESSIONAL SUMMARY

Senior Software Engineer with 10+ years of software engineering experience specializing in cloud-native platforms, Python, C++, SQL, and high-performance automated simulation environments. Proven track record of architecting scalable backend and cloud execution pipelines on Google Cloud Platform (GCP) and AWS, testing machine learning perception/planning models, and optimizing data layers. Experienced in building automated tools for simulation failure analysis and validating autonomous/robotics system software. Seeking a fully remote Software Engineer, Simulation role.

CORE COMPETENCIES

Languages & Core: Python, C++, SQL, TypeScript, JavaScript

Simulation & Systems: Simulation Environments, Autonomous System Software, Robotics Systems, Machine Learning & Computer Vision Integration, Failure Case Analysis

Cloud & DevOps: Google Cloud Platform (GCP), AWS (EKS, ECS, Lambda), Docker, Kubernetes, Terraform, CI/CD, GitHub Actions

Data & Architectures: Relational & Large-Scale Databases (PostgreSQL, SQL Server, MySQL), NoSQL (MongoDB, Redis), REST & GraphQL APIs, Microservices, Distributed Systems

PROFESSIONAL EXPERIENCE

Senior Software Engineer

Dec 2022 – Present | Rochester, NY

Netsmartz

- Built and optimized automated backend and simulation pipelines executing across Google Cloud Platform (GCP) and AWS infrastructure using Docker, Kubernetes, and Terraform.
- Designed and maintained modular software components in Python and C++ powering high-throughput data execution and model evaluation platforms.
- Integrated tools and scripts to evaluate machine learning perception and planning models; developed automated diagnostics to analyze simulation failure cases and improve model performance.
- Managed large-scale data layers across PostgreSQL, SQL Server, Redis, and vector stores, tuning complex SQL queries to reduce median API response time from 850ms to 380ms (32% performance boost).
- Led backend architecture decisions, code reviews, and mentored a team of 5 engineers while enforcing strict OAuth2, JWT, and RBAC security protocols.

Senior Full Stack Software Engineer

Feb 2020 – Nov 2022 | Chicago, IL

Tempus AI

- Developed and validated automated simulation environments and data pipelines in Python and C++ for large-scale computer vision and machine learning model validation.
- Executed complex SQL data extraction and large-scale computing tasks on AWS and GCP cloud environments, reducing p95 latency from 620ms to 470ms (24% reduction).
- Containerized and deployed backend and simulation components using Docker and Kubernetes to enable scalable software execution.

- Implemented robust security standards (OAuth2, JWT, RBAC, field-level encryption, audit logging) ensuring HIPAA and enterprise compliance across all services.

Software Engineer

Dec 2015 – Jan 2020

Talmate

- Developed scalable backend services and REST APIs using Java (Spring Boot) and Node.js to integrate complex enterprise workflows and third-party systems.
- Built memory-caching solutions using Redis and optimized relational queries on MySQL and Oracle, reducing workflow execution times from 12 minutes to 5.5 minutes (54% speedup).
- Designed automated CI/CD build scripts using Gradle, GitHub Actions, and Azure DevOps across containerized environments.
- Participated in architectural reviews and mentored team members on clean code standards and Agile methodologies.

EDUCATION

Bachelor's Degree in Computer Science

Aug 2011 – May 2015 | Houston, TX

University of Houston